## Query Search Performance

- **Query 1:** What are the criteria for a Member to be eligible for insurance?
    - *Screenshot 1:* Output of the search layer

```
==============================================================================
QUERY 1: What are the criteria for a Member to be eligible for insurance?
==============================================================================

****************************** TOP 3 RETRIEVED CHUNKS ******************************

--- Rank 1 ---
applicable premium rates in effect on the Policy Anniversary. This policy has been updated effective January 1, 2014 PART III - INDIVIDUAL REQUIREMENTS AND RIGHTS GC
6006 Section A - Eligibility, Page 1 PART III - INDIVIDUAL REQUIREMENTS AND RIGHTS Section A - Eligibility Article 1 - Member Life Insurance A person will be eligible
for Member Life Insurance on the date the person completes 30 consecutive days of continuous Active Work with the Policyholder as a Member. In no circumstance will a
person be eligible for Member Life Insurance under this Group Policy if the person is eligible under any other Group Term Life Insurance policy underwritten by The
Principal. Article 2 - Member Accidental Death and Dismemberment Insurance A person will be eligible for Member Accidental Death and Dismemberment Insurance on the
latest of: a. the date the person is eligible for Member Life Insurance; or b. the

--- Rank 2 ---
can be in force (see b. above and e. below). e. Proof of Good Health Requirements The type and form of required Proof of Good Health will be determined by The Principal.
 A Member must submit Proof of Good Health: (1) If ins urance for which a Member contributes a part of premium is requested more than 31 days after the date the Member
is eligible including any insurance the Member refuses and later requests. (2) If insurance is requested under this Group Policy by a Member that was eligible under the
Prior Policy, but elected to waive coverage under the Prior Policy. (3) If a Member has failed to provide required Proof of Good Health or has been refused insurance
under this Group Policy at any prior time. (4) If a Member elects to terminate insurance and, more than 31 days later, requests to be insured again. (5) If,

--- Rank 3 ---
only. No disabilit y or other benefits will be included. (3) The policy will be on one of the forms, other than term insurance, then issued by The Principal to persons
in the risk class to which the Member belongs on the individual policy's effective date. (4) Premium will be based on the Member's age and the standard rate of The
Principal for the policy form to be issued. b. Purchase Qualification A Member will qualify for individual purchase if insurance under this Group Policy terminates and:
(1) the Member's total Lif e Insurance, or any portion of it, terminates because he or she ends Active Work or ceases to be in a class eligible for insurance; or (2)
after the Member has been continuously insured under this Group Policy for at least five years, his or her total Memb er Life Insurance terminates because this Group
Policy terminates or
```

    - *Screenshot 2:* Final generated answer by gemini-2.0-flash.

```
****************************** FINAL GENERATED ANSWER******************************
<path_1>
Chunk 1 defines eligibility for Member Life Insurance and Member Accidental Death and Dismemberment Insurance. A person is eligible for Member Life Insurance after
completing 30 consecutive days of continuous Active Work with the Policyholder as a Member. They are not eligible if they are eligible under any other Group Term Life
Insurance policy underwritten by The Principal. A person is eligible for Member Accidental Death and Dismemberment Insurance on the later of the date they are eligible
for Member Life Insurance or the date specified in the policy.
</path_1>
<path_2>
Chunk 2 discusses Proof of Good Health requirements in several scenarios, such as when insurance is requested more than 31 days after eligibility, if coverage was waived
 previously, or if insurance was previously refused. This chunk does not directly define the initial eligibility criteria.
</path_3>
<path_3>
Chunk 3 describes the conditions under which a member can purchase an individual policy if their group insurance terminates. It does not define the initial eligibility
criteria for insurance.
</path_3>
<synthesis>
Chunk 1 provides the specific eligibility criteria for both Member Life Insurance and Member Accidental Death and Dismemberment Insurance. Chunks 2 and 3 are not
relevant to the initial eligibility requirements.
</synthesis>
<final_answer>
To be eligible for Member Life Insurance, a person must complete 30 consecutive days of continuous Active Work with the Policyholder as a Member and not be eligible
under any other Group Term Life Insurance policy underwritten by The Principal. For Member Accidental Death and Dismemberment Insurance, eligibility begins on the later
of the date they are eligible for Member Life Insurance.
</final_answer>
```

- **Query 2:** What happens if I fail to pay the premium before the grace period expires?

  - *Screenshot 3:* Output of the search layer.

```
================================================================================
QUERY 2: What happens if I fail to pay the premium before the grace period expires?
================================================================================

****************************** TOP 3 RETRIEVED CHUNKS ******************************

--- Rank 1 ---
total premium due has not been received by The Principal before the end of the Grace Period. Failure by the Policyholder to pay the premium within the Grace Period will
be deemed notice by the Policyholder to The Principal to discontinue this Group Policy at the end of the Grace Period. Article 2 - Termination Rights of the Policyholder
 The Policyholder may terminate this Group Policy effective on the day before any premium due date by giving Written notice to The Principal prior to that premium due
date. The Policyholder's issuance of a stop- payment order for any amounts used to pay premiums for the Policyholder's coverage will be considered Written notice from
the Policyholder. Article 3 - Termination Rights of The Principal The Principal may nonrenew or terminate this Group Policy by giving the Policyholder 31 days advance
notice in Writing, if the Policyholder: a. ceases to be actively

--- Rank 2 ---
an Insurance Month, premium for that Member will be adjusted and applied as if the change were to take place on the first of the next following Insurance Month. Article
5 - Contributions from Members Members are not required to contribute a part of the premium for their Member insurance under this Group Policy. Members are required to
contribute a part of the premium for their Dependent's insurance under this Group Policy. This policy has been updated effective January 1, 2014 PART II - POLICY
ADMINISTRATION GC 6005 Section C - Policy Termination, Page 1 Section C - Policy Termination Article 1 - Failure to Pay Premium This Group Policy will terminate at the
end of the Grace Period if total premium due has not been received by The Principal before the end of the Grace Period. Failure by the Policyholder to pay the premium
within the Grace Period will

--- Rank 3 ---
premium due date. The Group Policy will remain in fo rce until the end of the Grace Period, unless the Group Policy has been terminated by notice as described in PART
II, Section C. The Policyholder will be liable for payment of the premium for the time this Group Policy remains in force during the Grace Period. Article 2 - Premium
Rates The premium rate(s) for each Member insured for Life Insurance will be: a. Member Life Insurance $0.210 for each $1,000 of insurance in force. b. Member Accidental
 Death and Dismemberment Insurance $0.025 for each $1,000 of Member Life Insurance in force. c. Dependent Life Insurance $1.46 for each Member insured for Dependent Life
 Insurance. If the Policyholder has at least two other eligible group insurance policies underwritten by The Principal, as determined by The Principal, the Policyholder
may be eligible for a multiple policy discount. Article 3 -
```

  - *Screenshot 4:* Final generated answer.

```
****************************** FINAL GENERATED ANSWER******************************
<path_1>
Chunk 1 states that the Group Policy will terminate at the end of the Grace Period if the total premium due has not been received by The Principal before the end of the
Grace Period. It also mentions that failure to pay the premium within the Grace Period will be deemed notice by the Policyholder to The Principal to discontinue the
Group Policy at the end of the Grace Period.
</path_1>
<path_2>
Chunk 2 reiterates that the Group Policy will terminate at the end of the Grace Period if the total premium due has not been received by The Principal before the end of
the Grace Period.
</path_2>
<path_3>
Chunk 3 states that the Group Policy will remain in force until the end of the Grace Period.
</path_3>
<synthesis>
The information from all three chunks indicates that if the premium is not paid before the end of the grace period, the group policy will terminate.
</synthesis>
<final_answer>
If you fail to pay the premium before the grace period expires, the group policy will terminate.
</final_answer>
```

- **Query 3:** What constitutes 'Proof of Good Health' according to this policy?
    - *Screenshot 5:* Output of the search layer.

```
================================================================================
QUERY 3: What constitutes 'Proof of Good Health' according to this policy?
================================================================================

****************************** TOP 3 RETRIEVED CHUNKS ******************************

--- Rank 1 ---
can be in force (see b. above and e. below). e. Proof of Good Health Requirements The type and form of required Proof of Good Health will be determined by The Principal.
 A Member must submit Proof of Good Health: (1) If ins urance for which a Member contributes a part of premium is requested more than 31 days after the date the Member
is eligible including any insurance the Member refuses and later requests. (2) If insurance is requested under this Group Policy by a Member that was eligible under the
Prior Policy, but elected to waive coverage under the Prior Policy. (3) If a Member has failed to provide required Proof of Good Health or has been refused insurance
under this Group Policy at any prior time. (4) If a Member elects to terminate insurance and, more than 31 days later, requests to be insured again. (5) If,

--- Rank 2 ---
the same day of each following year. Policyholder The entity to whom this Group Policy is issued (see Title Page). Prior Policy The Group Term Life coverage of either:
a. the Policyholder; or b. a business entity which has been obtained by the Policyholder through a merger or acquisition; for which this Group Policy is a replacement.
Proof of Good Health Written evidence that a person is insurable under the underwriting standards of The Principal. This proof must be provided in a form satisfactory to
 The Principal. Qualifying Event A Qualifying Event for Accelerated Benefits is a medical condition, which would, in the absence of extensive or extraordinary medical
treatment; result in a dramatically limited life span. Such conditions may include, BUT ARE NOT LIMITED TO, one or more of the following: a. coronary artery disease
resulting in an acute infarction or requiring surgery; b. permanent neurological deficit resulting from

--- Rank 3 ---
whether or not the Member is Actively at Work. (2) A change in the Member's Scheduled Benefit amount because of a change in the Schedule of Insurance (as described in
PART IV, Section A) by amendment to this Group Policy for which Proof of Good Health is required (see e. above) will be effective on the later of: - the date the change
would have been effective if Proof of Good Health had not been required; or - the date Proof of Good Health is approved by The Principal. h. Effective Date for Benefit
Changes Due to Changes Requested by the Member (1) A change in a Member's Scheduled Benefit amount because of a request by the Member for which Proof of Good Health is
not required (see e . above) will normally be effective on the date of the request. However, if the Member is not Actively at Work
```

    - *Screenshot 6:* Final generated answer

```
****************************** FINAL GENERATED ANSWER******************************
<path_1>
Chunk 1 defines when Proof of Good Health is required under various circumstances related to insurance requests and eligibility, particularly when a member contributes
to the premium or requests insurance after a certain period. It also mentions that the type and form of required Proof of Good Health will be determined by The
Principal.
</path_1>
<path_2>
Chunk 2 provides a direct definition: "Proof of Good Health Written evidence that a person is insurable under the underwriting standards of The Principal. This proof
must be provided in a form satisfactory to The Principal." It also defines related terms like "Policyholder," "Prior Policy," and "Qualifying Event."
</path_3>
Chunk 3 discusses the effective dates for benefit changes, both those requiring and not requiring Proof of Good Health, and how these dates are affected by whether the
member is actively at work.
</path_3>
<synthesis>
Chunk 1 outlines the circumstances when Proof of Good Health is required, and Chunk 2 provides the definition of what Proof of Good Health actually is. Chunk 3 discusses
 effective dates for benefit changes related to Proof of Good Health. The most direct answer comes from Chunk 2.
</synthesis>
<final_answer>
According to this policy, Proof of Good Health is written evidence that a person is insurable under the underwriting standards of The Principal, and it must be provided
in a form satisfactory to The Principal.
</final_answer>
```

Test the Cache:

- If I send the same query again it should retrieve the chunks from the cache.

### 3.2 Test the Cache

Proving caching mechanism works instantly without re-embedding or querying ChromaDB.

```python
print("Testing Cache Execution...")
cache_test_query = queries[0]
print(f"Query: {cache_test_query}")
cached_chunks = search_and_rerank(cache_test_query)
```

```
Testing Cache Execution...
Query: What are the criteria for a Member to be eligible for insurance?
-> Result served from Cache!
```